Elevated CO_2 stimulates marsh elevation gain, counterbalancing sea-level rise

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Tidal wetlands experiencing increased rates of sea-level rise (SLR) must increase rates of soil elevation gain to avoid permanent conversion to open water. The maximal rate of SLR that these ecosystems can tolerate depends partly on mineral sediment deposition, but the accumulation of organic matter is equally important for many wetlands. Plant productivity drives organic matter dynamics and is sensitive to global change factors, such as rising atmospheric CO_2 concentration. It remains unknown how global change will influence organic mechanisms that determine future tidal wetland viability. Here, we present experimental evidence that plant response to elevated atmospheric [CO_2] stimulates biogenic mechanisms of elevation gain in a brackish marsh. Elevated CO_2 (ambient + 340 ppm) accelerated soil elevation gain by 3.9 mm yr^{-1} in this 2-year field study, an effect mediated by stimulation of below-ground plant productivity. Further, a companion greenhouse experiment revealed that the CO_2 effect was enhanced under salinity and flooding conditions likely to accompany future SLR. Our results indicate that by stimulating biogenic contributions to marsh elevation, increases in the greenhouse gas, CO_2, may paradoxically aid some coastal wetlands in counterbalancing rising seas.

coastal wetlands | nitrogen pollution | tidal marsh loss | root productivity | salinity

The world currently loses thousands of hectares of low-lying coastal wetlands to shallow open water each year (1–3), attributable, in part, to a recent acceleration of sea-level rise (SLR) (4–6). Loss of coastal wetlands threatens critical services these ecosystems provide, such as supporting commercially important fisheries, providing a wildlife habitat, improving water quality, and buffering human populations from oceanic forces (3). Recent catastrophes, such as Hurricane Katrina and the Asian Tsunami, have underscored the importance of understanding factors that govern sustainability of coastal wetlands in the face of climate change and accelerating SLR. Marshes must build vertically through accumulation of mineral and organic matter to maintain a constant elevation relative to sea level (7). To explain the dynamics of coastal wetland elevation, researchers have traditionally focused on abiotic factors, such as reductions of mineral sediment loads from hydrologic modifications (8). However, organic matter dynamics have a clear importance in peaty soils, which are composed mostly of live and dead plant tissues (9, 10) and may also play an important role in stabilizing mineral soils (11). Organic mechanisms may be especially sensitive to other global change factors and may determine the fate of tidal wetlands.

Rising atmospheric CO_2 is largely responsible for recent global warming and will continue to contribute to accelerating SLR through thermal expansion and ice melt (12). Elevated CO_2, in addition to accelerating SLR, may have important biologically mediated effects on coastal wetland ecosystems, such as stimulating plant productivity (13). The effects of elevated CO_2 must be considered along with other regionally or locally important factors that are changing simultaneously, such as nitrogen enrichment (14), flooding (15), and salinity (13).

We manipulated atmospheric CO_2 and soil nitrogen availability in a tidal marsh with low mineral sediment inputs, allowing us to isolate organic controls on marsh elevation in Chesapeake Bay. This experiment was designed to track sensitively how changes in plant productivity influenced soil elevation. Determining soil surface elevation change is important for 2 reasons. First, the loss in soil elevation relative to local sea level may provide an early indication of the syndrome of tidal wetland collapse (16). Second, tracking elevation changes in discrete strata through time, along with measurements of plant productivity and other environmental variables, allows identification of specific mechanisms critical to the persistence of tidal wetlands under accelerating SLR. To examine how CO_2 may interact with other factors that will accompany SLR, we manipulated CO_2, salinity, and flooding in a companion greenhouse study (17).

Results and Discussion

Elevated CO_2 stimulated above-ground productivity in the marsh by 30% in 2006 (Table 1; two-way ANOVA, CO_2 effect: $P < 0.05$) but not in 2007 ($P > 0.70$) during a severe drought. These responses were consistent with a 20-year record of elevated CO_2 treatment in a previous CO_2 study on the same marsh (13). Strikingly, elevated CO_2 accelerated the rate of soil elevation gain over 2 years of treatment. Least-squares linear trends revealed a slight loss of elevation in ambient CO_2 (−0.9 mm yr^{-1}) compared with an elevation gain (3.0 mm yr^{-1}) in the elevated CO_2 treatment (Fig. 1*A*; two-way ANOVA comparing linear trends, $n = 5$, CO_2 effect: $P < 0.05$). Nitrogen addition tended to affect elevation negatively, although the overall nitrogen effect was not significant. Deposition of mineral sediment on the marsh surface is negligible at this site. Root zone expansion by accumulation of plant material is essential to maintaining a constant surface elevation relative to rising sea level. Indeed, changes in marsh surface elevation mirrored changes in root zone thickness (Fig. 1*B*). Elevated CO_2 caused an increase in thickness of 4.9 mm yr^{-1} compared with 0.7 mm yr^{-1} in the ambient CO_2 treatment, an effect that could increase marsh tolerance for SLR.

Increases in soil elevation were driven by a stimulation of subsurface plant productivity. Changes in soil elevation over each growing season closely followed the pattern of treatment

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Table 1. Above-ground plant biomass and fine root production over 2 growing seasons of the field study

Treatments	Aboveground Jul-06	Aboveground Oct-06	Fine roots 2006	Aboveground July-07	Aboveground Oct-07	Fine roots 2007
Ambient CO_2	387 (27)[a]	677 (68)[a]	56 (11)[ab]	524 (53)[a]	449 (25)[a]	108 (17)[ab]
Ambient CO_2 +Nitrogen	603 (127)[b]	952 (144)[a]	40 (11)[a]	842 (149)[b]	697 (167)[b]	63 (9)[a]
Elevated CO_2	387 (26)[a]	883 (111)[a]	98 (39)[ab]	530 (39)[a]	399 (34)[a]	146 (14)[b]
Elevated CO_2 +Nitrogen	560 (44)[ab]	1393 (82)[b]	110 (13)[b]	902 (65)[b]	702 (55)[b]	104 (16)[ab]

Values represent means (SE) in g m^{-2}. Treatment means without common superscript symbols were different ($P \leq 0.05$) according to a Student's *t* test multiple comparison.

effects on fine root productivity (Fig. 2; $r^2 = 0.86$, $P < 0.05$). By the end of the second year, above-ground litter also may have entered the soil and contributed to a CO_2 effect on elevation. Rhizome and subsurface stem productivity were likely also important but could not be quantified nondestructively in the field. Although the presence or absence of an elevated CO_2 effect on above-ground production each year depended on climatic conditions, elevated CO_2 strongly stimulated fine root productivity both years, by 48% in 2006 (two-way ANOVA, $n = 5$, $P < 0.05$) and by 26% in 2007 ($P < 0.05$). This finding concurs with previous work at this site and many other CO_2 studies that report an inordinately large stimulation of root growth (18).

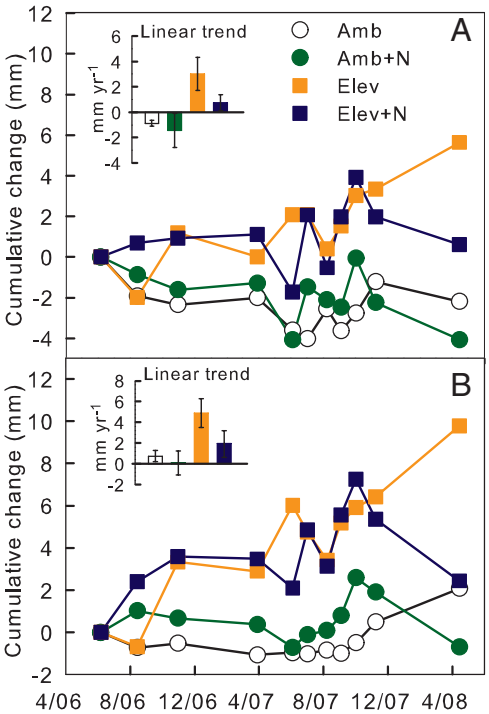

Fig. 1. Relative surface elevation (*A*) and root zone thickness (*B*) over 2 growing seasons in a brackish marsh (Chesapeake Bay) exposed to ambient (Amb) or elevated (Elev) atmospheric [CO_2] in open-top chambers and to factorial soil nitrogen (+N) enrichment. The slopes of the linear trends in elevation are inset in each panel (mean ± 1 SE). Elevated CO_2 increased elevation (two-way ANOVA, CO_2 effect: $P < 0.05$), an effect that mirrored CO_2 stimulation of root zone thickness ($P < 0.05$). Nitrogen addition had a nonsignificant tendency to reduce total elevation gain ($P = 0.18$) and root zone expansion ($P = 0.13$), despite stimulating above-ground plant productivity.

The timing of the below-ground response further supports the important role of plant growth in driving soil elevation gain. The CO_2 effects on root zone elevation arose during the first growing season (Fig. 1), concurrent with the production of live plant biomass. CO_2 effects on elevation during the growing season were not reversed by plant senescence and decomposition. Because anaerobic conditions in saturated soils substantially slow soil organic matter decay (11), the effects of productivity on elevation during the growing season persist through the nongrowing season. Consequently, elevated CO_2 forced the soil surface upward by stimulating the accumulation of both live and dead organic matter.

Other perturbations, particularly those that may accompany SLR, could modify the influence of elevated CO_2 on marsh elevation gain. To understand better how the CO_2 response under current sea-level conditions might differ with accelerating SLR, we examined the effect of CO_2 in combination with flooding and salinity in a mesocosm study. This experimental approach allowed isolation of root-zone controls on soil elevation dynamics and manipulation of factors known to influence root growth. The objective of our treatments was to vary the depth and duration of flooding in combination with a range of salinities anticipated under different SLR scenarios (17).

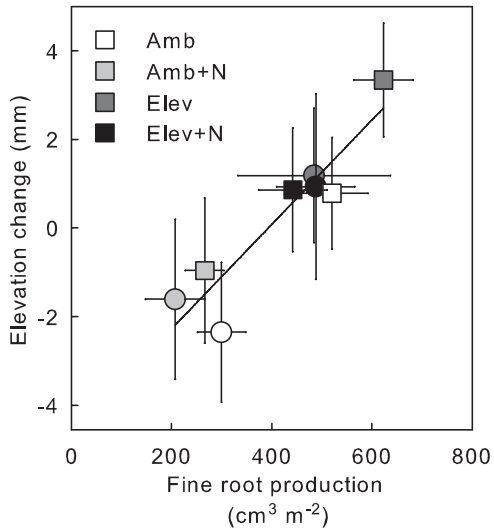

Fig. 2. Relation of subsurface biovolume production to vertical change in the field. Values represent treatment means (±1 SE) of relative elevation change and fine root productivity over each field season, 2006 (circles) and 2007 (squares). Amb, ambient; Elev, elevated; N, nitrogen.

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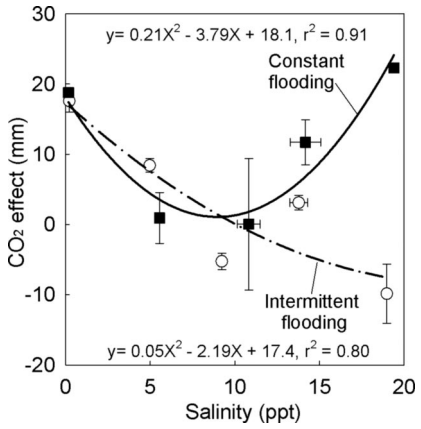

Fig. 3. Elevated CO_2 effects on marsh elevation in greenhouse mesocosms subjected to different salinity and flooding treatments. Values are the difference in elevation (mm) between mesocosms under elevated CO_2 (720 ppm) and ambient CO_2 (380 ppm). The *y* axis values represent mean differences in CO_2 response (± 1 SE) under intermittent (circles) and constant (squares) flooding in relation to mean interstitial salinity (± 1 SE) measured biweekly in mesocosms (*x* axis). Second-order polynomial curves were fitted to the data; *t* tests of quadratic coefficients indicated a significant change over salinity for both flooding treatments ($P < 0.05$). Patterns of response for the 2 flooding treatments were significantly different ($F = 3.03$, $P = 0.10$; analysis of covariance).

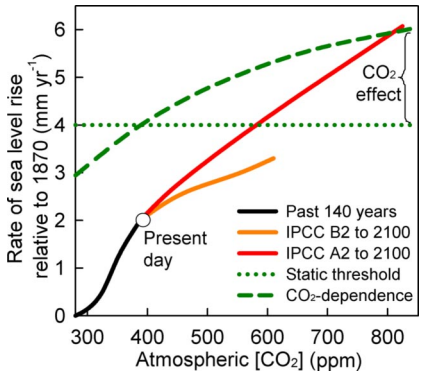

Fig. 4. Conceptual comparison of SLR and atmospheric $[CO_2]$ scenarios from the Inter-governmental Panel on Climate Change (IPCC) to hypothetical thresholds of marsh tolerance for increasing SLR. The black line approximates a simplified SLR × CO_2 relation for the past 140 years (4, 12, 29, 30). Red and orange lines represent most probable increases in atmospheric $[CO_2]$ and SLR rates for the years 2090–2100 according to model projections for 2 moderate IPCC socioeconomic scenarios, A2 and B2 (31). The difference between the 2 lines depends on socioeconomic uncertainties. Each model output has its own error based on physical uncertainties that are not shown here. The relation between SLR and CO_2 appears approximately linear in the 100-year projections, but other relations are possible. The dotted green line represents a threshold SLR rate fixed at 4 mm yr^{-1}, a rate that would threaten many marshes (6). The dashed green line represents a hypothetical threshold rate that depends on atmospheric $[CO_2]$ and is based on our finding of increased marsh elevation gain with elevated CO_2. The CO_2 effect (difference between the 2 green lines) portrayed here (≈2 mm yr^{-1} at 720 ppm) is conservative compared with that measured in the field experiment (3.9 mm yr^{-1}) because of uncertainties in extrapolating from short-term CO_2 effects on elevation to actual marsh thresholds. Marshes collapse when the rate of SLR (red or orange lines) exceeds the threshold of marsh tolerance for SLR (green lines). Thresholds may have been lower in the past when $[CO_2]$ was at preindustrial levels (≈280 ppm), because many plants likely exhibited reduced below-ground growth compared with today, and thus made smaller contributions to marsh elevation gain.

As found in the field study, CO_2 had a significant effect on marsh elevations in mesocosms (17). However, the patterns of soil expansion, driven by below-ground production, varied in a complex manner because of interactions of CO_2, flooding, and salinity (Fig. 3). Under intermittent flooding, CO_2 enhanced elevation at lower salinities but had no effect or a negative effect at higher salinities (Fig. 3). Under constant flooding, however, the pattern of CO_2 response was different, with expansion greatest both above and below the current salinity range for this marsh type (Fig. 3). In particular, CO_2 treatment increased elevation in flooded mesocosms at high salinities (i.e., conditions that simulated future SLR). Although the precise CO_2 response in nature will depend on many other factors in addition to salinity and flooding, our findings indicate that future CO_2 effects on elevation in Kirkpatrick Marsh will likely be modified by conditions that accompany SLR.

Agricultural and municipal runoff increases nitrogen (N) loading in estuaries, particularly those that drain densely populated watersheds. In the field study, nitrogen addition, simulating coastal eutrophication, tended to negate elevation gains caused by elevated CO_2 alone (Fig. 1*A*, *Inset*). Although nitrogen addition stimulated above-ground biomass, it decreased root productivity (Fig. 2 and Table 1). N addition commonly decreases carbon allocation to root productivity (19). Furthermore, N addition has been shown to stimulate decomposition in other nitrogen-limited peat-based wetlands (20). Nitrogen pollution may contribute to marsh loss (5), although the mechanisms have not been described. In our study, N addition tended to reduce elevation gain, partially by reducing root productivity, but also may have stimulated organic matter decomposition.

Plant community composition will modify ecosystem responses to elevated CO_2. Plants with the C_3 photosynthetic pathway typically exhibit a stronger CO_2 response than C_4 plants (13). The CO_2 effect may be particularly important for C_3-dominated marshes experiencing SLR, which includes many brackish marshes and mangrove swamps. In this marsh community, the responsive C_3 species is more flood-tolerant and less salt-tolerant than the C_4 species (21, 22). The mesocosm experiment also provided insight into C_3 vs. C_4 contributions to soil expansion (17). Separation of roots by species was not possible, but species contributions to soil volume in the mesocosms could be assessed by measuring the final volume of shoot bases (i.e., the below-ground portion of the shoot that serves as a storage organ in graminoids). Vertical change in soil elevation was positively correlated with subsurface shoot volume of the C_3 but not the C_4 species (17). Consequently, the potential for CO_2 stimulation under an accelerating SLR scenario may depend on the inherent tolerance of component species to salinity, flooding, and other factors as well as their potential to respond to CO_2 (17).

Our findings bear particular importance given the threat of accelerating SLR to coastal wetlands worldwide. A recent report suggests that a 2-mm increase in the rate of SLR will threaten or eliminate a large portion of mid-Atlantic marshes (6). The threshold rate of SLR (i.e., the SLR rate at which the marsh converts to open water) depends on the rate of marsh elevation gain by accumulating sediment or endogenous organic material (7). For instance, models predict that increasing mineral sediment load should increase this threshold rate (7, 23). Nevertheless, many marshes, like Kirkpatrick Marsh, naturally receive very little exogenous sediment. Therefore, the critical SLR threshold depends almost entirely on biological mechanisms that have been more difficult to describe and predict. We suggest that elevated CO_2, by stimulating below-ground plant productivity and biogenic elevation gain, may also increase the capacity of coastal wetlands to tolerate SLR (Fig. 4).

Atmospheric CO_2 has risen 40% from preindustrial levels (≈280 ppm) to the current concentration (≈388 ppm) over the past 200 years and has enhanced plant productivity globally. A CO_2 effect on coastal wetland elevation may have already

mitigated a significant amount of marsh loss acting through the biological mechanisms demonstrated in our study (Fig. 4). However, the stimulatory effects of CO_2 on leaf-level photosynthesis are known to diminish with each marginal increase in CO_2 concentration (24), and a similar attenuation of the stimulatory effect of elevated CO_2 on soil elevation gain should follow. Consequently, further increases in atmospheric CO_2 beyond 720 ppm may yield diminishing benefits for plant productivity and marsh elevation, while further accelerating SLR through greenhouse warming (12). A simultaneous increase in plant stressors, such as salinity and flooding, however, may sustain or enhance the CO_2 effect, as found in the mesocosm study.

Our findings show that elevated CO_2 stimulates plant productivity, particularly below ground, thereby boosting marsh surface elevation. In addition to providing evidence for organic control of wetland topography on the landscape scale, we show how elevated CO_2 may strengthen the organic mechanisms that sustain marsh elevation relative to SLR in the future. These effects likely extend beyond biogenic marshes, because organic processes can exert control over elevation dynamics in mineral-soil wetlands as well (11). For instance, marshes in the Mississippi River delta have high rates of subsidence that were naturally countered by mineral sediment deposition (25). However, hydrological modifications by humans have diminished sediment deposition in this system. Such sediment-deficient deltas, which once would have been buffered against increasing SLR, have been rendered more vulnerable, and hence more dependent on organic contributions to sustain a constant elevation relative to SLR. To manage or restore such ecosystems to a self-sustaining state, we must explore further how global change interactions will influence key biological mechanisms that maintain soil elevations in coastal wetland ecosystems.

Materials and Methods

The field study was conducted in Kirkpatrick Marsh (38°53′ North, 76°33′ West), located on a microtidal subestuary of Chesapeake Bay. The site is dominated by a perennial C_3 sedge, *Schoenoplectus americanus*, and the C_4 perennial grasses *Spartina patens* and *Distichlis spicata* and is representative of brackish high marshes of mid-Atlantic North America (26). The soils at the site are organic (>80% organic matter) to a depth of ≈5 m. Mean tidal range is 44 cm. The high marsh zone is 40–60 cm (26) above daily mean low water level and is inundated by 28% of high tides. Salinity averages 10 parts per thousand (ppt) and ranges seasonally from 4 to 15 ppt. The average daily air temperature reaches a low of −4 °C in January and a high of 31 °C in July. The 80-year trend of SLR is ≈3.4 mm yr^{-1} (tidesandcurrents.noaa.gov, Annapolis), which outpaces the long-term local eustatic trend attributable to glacial isostatic adjustment (27).

Beginning in May 2006, we exposed 20 3.3-m^2 marsh plots to 2 atmospheric CO_2 concentrations (ambient and ambient + 340 ppm) to simulate a likely atmospheric CO_2 concentration projected for the year 2100. The 2 CO_2 levels were crossed with 2 levels of N addition (0 and 25 g of nitrogen m^{-2} yr^{-1}). Ammonium chloride was dissolved in 5 L of water from the Rhode River, the subestuary adjacent to the site. On 5 dates (approximately monthly, avoiding high tides) throughout the growing season, we used backpack sprayers to deliver a fertilizer solution to 10 plots. The fertilizer solution was then rinsed from standing vegetation with another 5 L of unamended river water applied with backpack sprayers. Each fertilizer application simulated 5 g of nitrogen m^{-2} in the equivalent of 0.5 cm of river water. The 10 unfertilized chambers received 10 L of unamended river water applied in the same manner (see *SI Text*).

We outfitted each plot with a surface elevation table (SET) to measure soil elevation change sensitively (Figs. S1 and S2). Deep-rod SETs, used in conjunction with shallow benchmarks, enabled elevation changes to be partitioned between the root zone (0–0.3 m in depth) and the deep zone (0.3–4.5 m in depth). We used elevation measurements in adjacent unchambered reference plots to account for chamber effects and to minimize the confounding effects of spatial variability.

Greenhouse Methods

To examine how CO_2 might affect elevation response under simulated SLR scenarios, data from a greenhouse mesocosm experiment (17) were used. This experiment used the same marsh type as the field study at the mid-Atlantic location but was conducted using soil and plant material from the Mississippi River delta and carried out at the Wetland Elevated CO_2 Experimental Facility (National Wetlands Research Center, Lafayette, LA). A detailed description of the facility and the experimental design is given elsewhere (17, 28) but is briefly repeated here to provide context for this study.

Blocks of peat with intact vegetation (sods) were collected from a brackish marsh community (*ca.* 50:50 mixture of *S. americanus* and *S. patens*) in coastal Louisiana and established in mesocosms (0.1-m^2 area × 0.4 m in depth, 25-L volume). In March 2005, 60 replicate mesocosms were randomly assigned to 1 of 4 greenhouses with atmospheric concentrations of 380 (ambient) or 720 (elevated) ppm of CO_2, 3 flooding levels (constantly flooded, intermittently flooded, drained), and 5 salinity levels (0, 5, 10, 15, 20 ppt of sea salts). All above-ground material was removed on senescence so that the only inputs were from below-ground production. Expansion or contraction of the marsh sods was measured for 1 year with mini-SETs [modeled after SETs used in the field study (17)]. CO_2 response was calculated as the difference between elevated and ambient CO_2 treatments and plotted vs. interstitial salinity (averaged over experimental period).

The drained treatment was not included in the analysis, because our objective in this study was to understand better how CO_2 response to current SLR (observed in the field) might change under a future SLR scenario of increased flooding. Second-order polynomial curves were fitted to the data, and overall differences were tested with analysis of covariance (using salinity as a continuous effect and flooding as a categorical effect in the model); *t* tests of quadratic and linear coefficients (for each fitted curve) also were used to assess differences.

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1. Dahl TE (2005) *Status and Trends of Wetlands in the Conterminous United States, 1998 to 2004* (U.S. Department of the Interior, Fish and Wildlife Service, Washington, DC).
2. Valiela I, Bowen JL, York JK (2001) Mangrove forests: One of the world's threatened major tropical environments. *Bioscience* 51:807–815.
3. Zedler JB, Kercher S (2005) Wetland resources: Status, trends, ecosystem services, and restorability. *Annual Review of Environment and Resources* 30:39–74.
4. Church JA, White NJ (2006) A 20th century acceleration in global sea-level rise. *Geophys Res Lett* 33, 10.1029/2007GL029703.
5. Hartig EK, Gornitz V, Kolker A, Mushacke F, Fallon D (2002) Anthropogenic and climate-change impacts on salt marshes of Jamaica Bay, New York City. *Wetlands* 22:71–89.
6. Reed DJ, et al. (2008) *Site-Specific Scenarios for Wetlands Accretion as Sea Level Rises in the Mid-Atlantic Region. Section 2.1. Background Documents Supporting Climate Change Science Program Synthesis and Assessment Product*, eds Titus JG, Strange EM (EPA 430R07004, U.S. EPA, Washington, DC).
7. Morris JT, Sundareshwar PV, Nietch CT, Kjerfve B, Cahoon DR (2002) Responses of coastal wetlands to rising sea level. *Ecology* 83:2869–2877.
8. Turner RE (2004) Coastal wetland subsidence arising from local hydrologic manipulations. *Estuaries* 27:265–272.
9. Nyman JA, Chabreck RH, Kinler NW (1993) Some effects of herbivory and 30 years of weir management on emergent vegetation in brackish marsh. *Wetlands* 13:165–175.
10. McKee KL, Cahoon DR, Feller IC (2007) Caribbean mangroves adjust to rising sea level through biotic controls on change in soil elevation. *Global Ecol Biogeogr* 16:545–556.
11. Nyman JA, Walters RJ, Delaune RD, Patrick WH (2006) Marsh vertical accretion via vegetative growth. *Estuar Coast Shelf Sci* 69:370–380.
12. Bindoff NL, et al. (2007) *Climate Change 2007: The Physical Science. Basis Contribution of Working Group I to the Fourth Assessment Report of the Intergovernmental Panel on Climate Change*, eds Solomon S, et al. (Cambridge Univ Press, New York), pp 387–429.
13. Erickson JE, Megonigal JP, Peresta G, Drake BG (2007) Salinity and sea level mediate elevated CO_2 effects on C-3-C-4 plant interactions and tissue nitrogen in a Chesapeake Bay tidal wetland. *Global Change Biol* 13:202–215.

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14. Morris JT, Bradley PM (1999) Effects of nutrient loading on the carbon balance of coastal wetland sediments. *Limnol Oceanogr* 44:699–702.
15. Donnelly JP, Bertness MD (2001) Rapid shoreward encroachment of salt marsh cordgrass in response to accelerated sea-level rise. *Proc Natl Acad Sci USA* 98:14218–14223.
16. Cahoon DR, et al. (2003) Mass tree mortality leads to mangrove peat collapse at Bay Islands, Honduras after Hurricane Mitch. *J Ecol* 91:1093–1105.
17. Cherry JA, McKee K, Grace JB (2009) Elevated CO_2 enhances biological contributions to elevation change in coastal wetlands by offsetting stressors associated with sea-level rise. *J Ecol* 97:67–77.
18. Rogers HH, Runion GB, Krupa SV (1994) Plant-responses to atmospheric CO_2 enrichment with emphasis on roots and the rhizosphere. *Environ Pollut* 83:155–189.
19. Agren GI, Franklin O (2003) Root:shoot ratios, optimization and nitrogen productivity. *Ann Bot (London)* 92:795–800.
20. Bragazza L, et al. (2006) Atmospheric nitrogen deposition promotes carbon loss from peat bogs. *Proc Natl Acad Sci USA* 103:19386–19389.
21. Broome SW, Mendelssohn IA, McKee KL (1995) Relative growth of *Spartina-patens* (Ait) Muhl and *Scirpus-olneyi* Gray occurring in a mixed stand as affected by salinity and flooding depth. *Wetlands* 15:20–30.
22. Rasse DP, Peresta G, Drake BG (2005) Seventeen years of elevated CO_2 exposure in a Chesapeake Bay wetland: Sustained but contrasting responses of plant growth and CO_2 uptake. *Global Change Biol* 11:369–377.
23. Kirwan ML, Murray AB (2007) A coupled geomorphic and ecological model of tidal marsh evolution. *Proc Natl Acad Sci USA* 104:6118–6122.
24. Jacob J, Greitner C, Drake BG (1995) Acclimation of photosynthesis in relation to Rubisco and non-structural carbohydrate contents and *in situ* carboxylase activity in *Scirpus olneyi* grown at elevated CO_2 in the field. *Plant Cell Environ* 18:875–884.
25. Coleman JM, Roberts HH, Stone GW (1998) Mississippi River delta: An overview. *J Coast Res* 14:698–716.
26. Jordan TE, Correll DL (1991) Continuous automated sampling of tidal exchanges of nutrients by brackish marshes. *Estuar Coast Shelf Sci* 32:527–545.
27. Douglas BC (2005) Gulf of Mexico and Atlantic coast sea level change. *Geophysical Monograph* 161:111–121.
28. McKee K, Rooth JE (2008) Where temperate meets tropical: Multi-factorial effects of elevated CO_2, nitrogen enrichment, and competition on a mangrove-salt marsh community. *Global Change Biol* 14:971–984.
29. Gehrels WR, et al. (2005) Onset of recent rapid sea-level rise in the western Atlantic Ocean. *Quaternary Science Reviews* 24:2083–2100.
30. Donnelly JP, Cleary P, Newby P, Ettinger R (2004) Coupling instrumental and geological records of sea-level change: Evidence from southern New England of an increase in the rate of sea-level rise in the late 19th century. *Geophys Res Lett* 10.1029/2003GL 018933.
31. Meehl GA, et al. (2007) *Climate Change 2007: The Physical Science Basis. Contribution of Working Group I to the Fourth Assessment Report of the Intergovernmental Panel on Climate Change*, eds. Solomon S, et al. (Cambridge Univ Press, New York), pp 749–844.

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